A Dual-Stage Wavelet and Linear Attention Enhancement Network for Agricultural Hyperspectral Image Classification

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***Abstract*—Hyperspectral image (HSI) classification faces unique challenges in an agricultural scenario due to spectral–spatial feature similarity caused by complex planting structures and high spectral similarity. Existing spatial–spectral joint feature extraction methods fail to fully exploit the advantages of spatial and spectral information and thus have certain limitations and cannot effectively distinguish similar crops in agricultural scenarios. To address these limitations, we proposed a dual-stage wavelet and linear attention enhancement network (DSW-LAN) for agricultural HSI classification, addressing the challenges of complex spatial–spectral information and high redundancy. We integrate a direction factorized deformable 3-D convolution (DFDWConv3D) module to capture multiscale spatial–spectral features through adaptive kernel adjustments, while the wavelet transform decomposes spatial features into low-frequency (LL) (structural) and high-frequency (textural) components for targeted enhancement. Additionally, a spectral probe-guided linear attention mechanism efficiently models long-range spectral dependencies with reduced computational complexity by prioritizing discriminative bands. Experimental results demonstrate superior performance on three challenging agricultural HSI datasets, achieving enhanced classification accuracy with reduced computational complexity.**

***Index Terms*—Agricultural hyperspectral image (HSI) classification, factorized kernel, spatial–spectral feature extraction, spectral attention mechanism, wavelet transform.**

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I. Introduction

HYPERSPECTRAL images are a class of 3-D images that capture rich spectral and spatial information, typically spanning hundreds of spectral bands, enabling the identification of material-specific spectral signatures on object surfaces. Hyperspectral image (HSI) classification has become crucial in precision agriculture for tasks such as crop species discrimination, health monitoring, yield estimation, and pest/disease early warning. While deep learning approaches have surpassed traditional statistical methods, agricultural HSI classification faces unique challenges such as mixed pixels, spectral resolution limitations, and spectral variability, all of which contribute to suboptimal performance.

Traditional machine learning in HSI classification centers on handcrafted spectral–spatial features and extracts key information through dimensionality reduction and feature fusion. Widely used methods include classical algorithms such as principal component analysis (PCA) [1], linear discriminant analysis (LDA) [2], locally linear embedding (LLE) [3], as well as newer techniques like band selection [4]. Then, researchers employ classifiers like support vector machines (SVMs) [5], random forests [6], and k-nearest neighbors (k-NNs) [7] for efficient classification. It offers advantages such as low computational cost, strong adaptability to small-sample scenarios, and high interpretability. For instance, Guo et al. [8] introduced a spectral-weighted kernel approach for SVM in HSI classification, which enhances the performance by adaptively optimizing band-specific weights. Xia et al. [9] proposed the RoRF-KPCA method that randomly partitioned the feature space into subsets for nonlinear high-order feature extraction via kernel PCA, fusing the transformed features through a random forest classifier with majority voting. Tu et al. [10] introduced a superpixel-based framework with dual k-selection rules, where k1 representative training samples and k2 test pixels per superpixel are selected using k-nearest neighbors to address intraclass variability and interclass interference. However, traditional machine learning methods have a limited ability to model complex nonlinear

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relationships. It is suitable for scenarios with limited labeled data or constrained resources, and its performance is often enhanced through multifeature fusion or integration with deep learning techniques.

With the breakthrough advancements in deep learning, data-driven feature learning approaches have significantly improved HSI classification accuracy. Researchers have proposed several novel deep learning architectures. Among these, convolutional neural networks (CNNs) have become mainstream owing to their local receptive fields and weight-sharing properties, which excel at spatial feature extraction. However, they often struggle with precise spectral feature extraction. To address this limitation, Lei et al. [11] proposed a graph angle field-based Siamese graph convolutional network, which extracts deep features for individual pixels while enhancing interpixel discriminability through superpixel-based spectral–spatial information fusion. Traditional CNNs rely on fixed-shape convolutional kernels, which fail to capture long-range dependencies among HSI pixels. To overcome this, Fu et al. [12] modified the convolutional block attention module, enabling dynamic adjustment of receptive field sizes based on multiscale HSI information, and jointly emphasizing critical spectral and spatial features. Yang et al. [13] proposed DMSGer, which constructs region-level graphs via superpixel segmentation and metric learning, followed by dynamic pixel-level feature updates to refine representations. Although group convolution networks show promise in HSI classification, they often neglect interband spectral correlations. Duan et al. [14] addressed this problem with a group interaction threshold attention network, which utilizes a hierarchical split-and-cascade strategy to propagate inter-group information and incorporates learnable threshold blocks to assess pixel-wise spectral validity across diverse spectral ranges. Ashraf et al. [15] proposed an improved 3-D UNet architecture with a modified convolutional neural network that integrates spatial–spectral information to address challenges such as nonlinear pixel combinations and limited labeled data.

The advent of Transformer has revolutionized HSI classification by compensating for CNNs' limitations in long-range dependency modeling. Song et al. [16] introduced ISSFormer, which jointly learns global spectral and local spatial features. Additionally, it introduces a bidirectional interaction mechanism, enabling interactions between local spatial features and global spectral features. Jia et al. [17] proposed CenterFormer, employing a center spatial–spectral attention mechanism to amplify central pixel representations and a multigranularity feature extractor to mitigate heterogeneous pixel interference. Fang et al. [18] developed a patch confidence-enhanced Transformer, which dynamically prioritizes informative regions via a patch quality assessment module and optimizes global context through spectral–spatial multifeature fusion and an enhanced additive attention Transformer.

Hybrid CNN–Transformer architectures have gained increasing attention for their ability to synergize local and global feature extraction. Ding et al. [19] proposed a spectral–spatial convolution-enhanced long-range dependency Transformer network, which dynamically models local features and aggregates multiscale contextual relationships to capture comprehensive HSI characteristics. Fang et al. [20] developed the SClusterFormer, integrating deformable convolutions with spectral–spatial cluster-based attention mechanisms to enhance fine-grained details and structural representations, effectively capturing both high-frequency anomalies and low-frequency (LL) trends while optimizing outputs through pixel-level cross-feature fusion. Yang et al. [21] introduced a parallel hybrid framework where a CNN branch extracts local contextual features, while a Transformer branch captures global dependencies, adaptively fusing these features across multiple resolutions. Wu et al. [22] designed a multiscale spatial–spectral shuffling convolution with 3-D lightweight Transformer, which combines multiscale spatial–spectral shuffling operations and a lightweight 3-D Transformer to preserve 3-D structural features throughout the feature extraction pipeline while maintaining computational efficiency.

Recently, the selective state-space model (SSM) (Mamba) [23] has been introduced into the field of HSI classification, enabling long-range information modeling with linear complexity. The properties of Mamba significantly reduce the computational burden associated with extracting global spatial information, facilitating precise modeling of complex spatial structures. Ming et al. [24] introduced a semantic-tokenized Mamba model, which captures relationships among different semantic tokens through spectral–spatial joint feature extraction, semantic token sequence generation, and a semantic token state-space model. To address local detail loss in Mamba, Pan et al. [25] proposed MambaLG, which integrates SpaM for preserving 2-D spatial structure while capturing global semantics and SpeM for precise short- and long-range spectral modeling.

With the ongoing development of deep learning, researchers have begun integrating mathematical approaches to enhance the model performance. For instance, Ahmad et al. [26] proposed WaveMamba, which integrates wavelet transformation with the spatial–spectral Mamba architecture to capture local texture patterns and global contextual relationships. To address the challenges of preserving high-frequency details, Shibwabo et al. [27] proposed MWGN-SSA, which integrates a multiscale learnable wavelet network for frequency-domain decomposition, retaining critical high-frequency textures for small or spectrally subtle targets, and uses a window-based spectral self-attention for local–global spectral correlations. Singh and Bruzzone [28] proposed WIANet, which integrates wavelet convolution and attention mechanisms into a UNet-based architecture to exploit spectral-texture information for distinguishing spectrally similar land cover classes.

At the same time, cutting-edge breakthroughs in remote sensing provide profound universal insights for HSI classification tasks. Li et al. [29] propose a context-aware cascade cognition framework to address long-range context bottlenecks in satellite scene graphs, demonstrating analogous cross-band dependency modeling needs in farmland HSI classification that require efficient nonadjacent spectral channel correlation capture. Concurrently, Li et al. [30] proposed MFVNet that addresses the inherent resolution-receptive field conflict through multifield-of-view adaptive fusion, resolving the acute tradeoff between spatial detail and spectral coverage in hyperspectral classification tasks.

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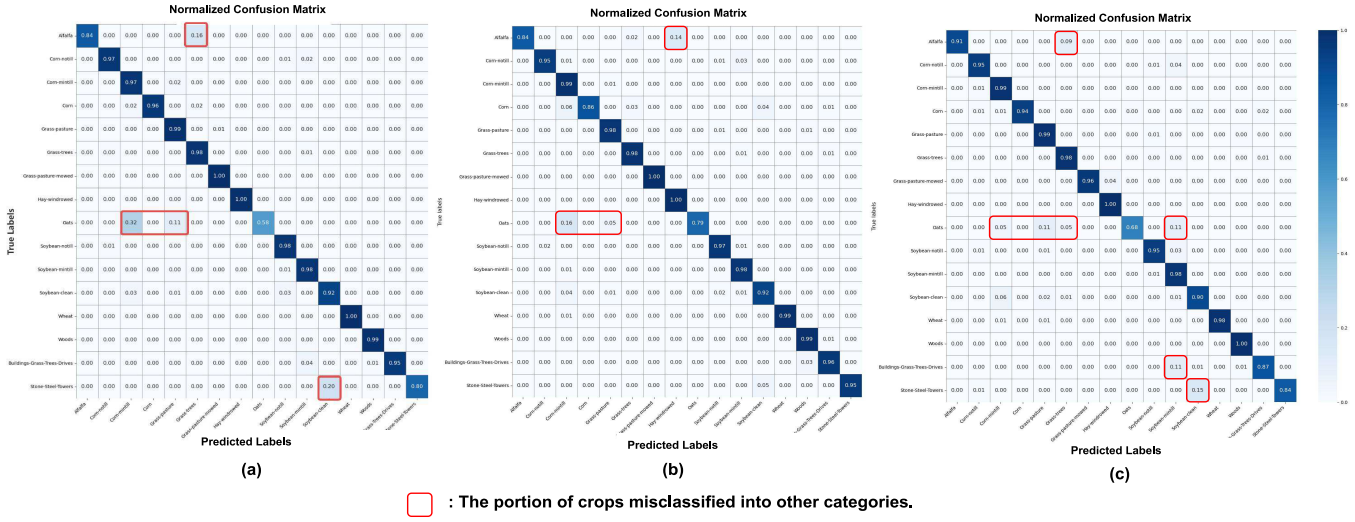


Fig. 1. Confusion matrix for the classification results on the Indian Pines dataset. (a) Morphformer. (b) IGroupSS-Mamba. (c) MSFI-CNet. The red boxes indicate the proportion of samples misclassified into other categories for each class, while the diagonal entries represent the proportion of correctly classified samples. Darker colors correspond to higher proportions.

Furthermore, Li et al. [31] achieved the fine-grained discrimination breakthrough by MEET through scene-in-scene architecture—whose core value lies in hierarchical context enhancement—can be directly transferred to agricultural HSI classification scenarios. Moreover, Wu et al. [32] demonstrated how progressive pretraining strategies with semantic enhancement can provide strong general-purpose representations, significantly advancing diverse Earth observation tasks. This work establishes an important benchmark for the community by showing the power of foundation models in capturing cross-modal knowledge and enabling few-shot generalization. Building upon these inspiring advances, our research specializes in the task of agricultural HSI classification, where unique challenges—such as subtle crop physiological variations, high spectral–spatial similarity, and redundant spectral information—demand tailored solutions.

More specifically, HSI classification in agricultural contexts presents unique challenges due to the subtle physiological variations in crops and complex planting patterns. These characteristics induce high spectral–spatial similarity among categories, leading to feature redundancy, blurred interclass boundaries, and increased computational complexity during joint spectral–spatial feature extraction. While existing spectral–spatial networks have made progress, they remain insufficient in addressing agriculture-specific issues, particularly in differentiating morphologically similar crop types. As demonstrated in Fig. 1, even advanced models like Morphformer [33] struggle to distinguish between highly similar categories such as grass-pasture-mowed and grass-pasture, highlighting the need for more discriminative feature learning mechanisms tailored to agricultural scenarios.

To address these challenges and overcome the limitations of existing networks, we proposed a dual-stage wavelet and linear attention enhancement network that is specifically designed for agricultural scenarios. The model divides feature learning into two stages. The first stage is an improved spectral–spatial joint feature extraction module, and the second stage focuses on enhancing spectral and spatial features. In the first stage, we employ the direction factorized deformable 3-D convolution (DFDWConv3D) block, which uses factorized kernels in different axial directions to extract multidirectional spectral–spatial features and cross-band spectral dependencies. Next, in the second stage, we apply wavelet-based convolution to transform the extracted features into the frequency domain, decomposing spatial features into LL and high-frequency (LH, HL, and HH) subbands to further enhance spatial feature representation. Unlike existing wavelet methods, our approach pioneers adaptive multilevel directional subband fusion within the feature enhancement stage, dynamically preserving crop structural contours while amplifying discriminative textures through learnable subband weighting. Finally, leveraging the 1-D nature of spectral information, we designed a spectral probe-guided linear attention mechanism that efficiently captures global spectral dependencies. Diverging from conventional quadratic-complexity attention or heuristic linear approximations, our spectral probe enables targeted band selection with strictly linear computational costs, maintaining interpretable attention distributions essential for agricultural spectral analysis. Unlike foundation models that aim for broad generalization across modalities, our proposed DSW-LAN is specifically tailored for agricultural HSI data. While foundation models provide a strong universal basis, our method complements them by offering a targeted and specialized solution to enhance classification performance in the agricultural domain.

The contribution of this article can mainly be summarized in the following.

1) We designed the DFDWConv3D block, which extracts local spectral–spatial features, captures broad spatial context, and models cross-band spectral dependencies, effectively leveraging spatial and spectral information to optimize spatial–spectral feature extraction.

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2) We designed a wavelet-domain spatial feature enhancement module that separates primary features into low- and high-frequency spatial components, isolating complex spatial patterns, which enhances edge and texture sensitivity, improving spatial feature discriminability.
3) We designed a linear-complexity attention mechanism guided by a 1-D spectral probe. It constructs global spectral dependencies and enhances discriminative bands while significantly reducing computational cost.

II. Related Works

In this section, we will introduce the most relevant research developments, including methods based on spatial–spectral feature extraction, frequency-domain transformation, and attention mechanisms. This provides a foundation for understanding the design and rationale behind our proposed modules.

A. Methods Based on Spatial–Spectral Joint Feature Extraction

The 3-D CNN is widely used due to its ability to extract spatial and spectral features simultaneously. Song et al. [34] proposed a few-shot HSI change detection method based on a 3-D convolution network, which utilizes 3-D convolution neural networks to extract spectral–spatial features while preventing overfitting, thus mitigating the issue of limited training samples. Esmaeili et al. [35] presented ResMorCNN, a model that combines 3-D convolutional layers with mathematical morphology to extract structural information, shapes, and interregional interactions from HSIs. To achieve enhanced integration of spatial–spectral features, researchers have turned to attention mechanisms and Transformer-based architectures in recent studies. Chhapariya et al. [36] combined spectral residual attention and spatial residual attention addressing high-dimensionality overfitting and spatial ambiguity in HSI classification while avoiding dimensionality reduction. Ahmad et al. [37] integrated a spatial–spectral Transformer with uncertainty-diversity querying and dynamic layer freezing to optimize HSI classification under limited labeled data. Li et al. [38] integrated a 3-D CNN-based hierarchical feature fusion network with a spectral–spatial Transformer to capture both local discriminative features and global pixel dependencies. Moreover, in remote sensing image (RSI) classification, MFVNet [30] employs a deep adaptive fusion approach that utilizes pyramid sampling to effectively handle the challenges of large field-of-views in RSI semantic segmentation. In fine-grained geospatial scene classification, Li et al. [31] introduced the context-aware Transformer model that employs adaptive fusion of spatial context to effectively classify fine-grained geospatial scene samples from the million-scale dataset.

The DFDWConv3D module we proposed further advances these concepts with our own improvements. Compared to RSI tasks, HSIs, which can be seen as 3-D structures, possess more complex spatial–spectral features, making it more challenging to extract and utilize local and global information. Our method adaptively models local spectral–spatial structures while efficiently capturing long-range dependencies across bands. By employing factorized kernels and depthwise separable convolutions, it significantly reduces computational costs while effectively extracting rich and spectral–spatial information.

B. Methods Based on Frequency-Domain Transformation

In an agricultural scenario, high class similarity and complex distributions make it difficult to extract discriminative spatial features, leading to challenges in boundary classification and limitations in traditional methods. Wavelet and Fourier transforms can address this by mapping spatial features into the frequency domain, effectively separating LL contours from high-frequency textures. Ahmad et al. [39] integrated the wavelet transform with a Transformer architecture to decompress feature maps without loss, enhancing structural and morphological interactions between image patches and spectral channels through wavelet decomposition. Ullah et al. [40] proposed the deep hyperspectral snapshot network, embedding wavelet transform into a convolutional architecture for low-complexity spectral analysis. Seydi et al. [41] introduced a wavelet-based network, using wavelet functions as learnable activation functions to nonlinearly map input spectral features, capturing multiscale spatial and spectral patterns through scale expansion and position translation. Xu et al. [42] proposed a 3-D wavelet convolutional neural network, which integrates 3-D discrete wavelet transform with CNNs to reduce computational complexity while enhancing feature discrimination through joint frequency-spatial modeling. Chen et al. [43] proposed a frequency-spatial–spectral prototype network, leveraging the fractional Fourier transform to enhance inter-class separability while maintaining consistency between classes.

Building on these concepts, our proposed network employs a wavelet transform to transform spatial information into the frequency domain, separating it into low- and high-frequency subbands for targeted feature extraction. This approach effectively overcomes the challenge of extracting discriminative features from overlapping frequency components.

C. Methods Based on Attention Mechanism

The attention mechanism dynamically allocates band weights through spectral channel attention, enhancing discriminative spectral responses while suppressing redundancy. With strong global modeling capabilities, it has shown excellent performance in HSI classification. Zhang et al. [44] designed a cascaded spatial cross-attention module that first applies cross-attention to local and global spatial features and then propagates important spatial regions across channels using a group cascade structure, ultimately obtaining joint attention features to improve network robustness. However, traditional attention mechanisms, despite their strong representation power, suffer from high computational costs $O(N^2)$.

Faced with computational efficiency bottlenecks, researchers have actively begun exploring ways to significantly reduce the computational complexity of attention mechanisms while maximizing the preservation of their powerful modeling capabilities. Linear attention, as a critical solution, has emerged to address this challenge. Babiloni et al. [45] demonstrated

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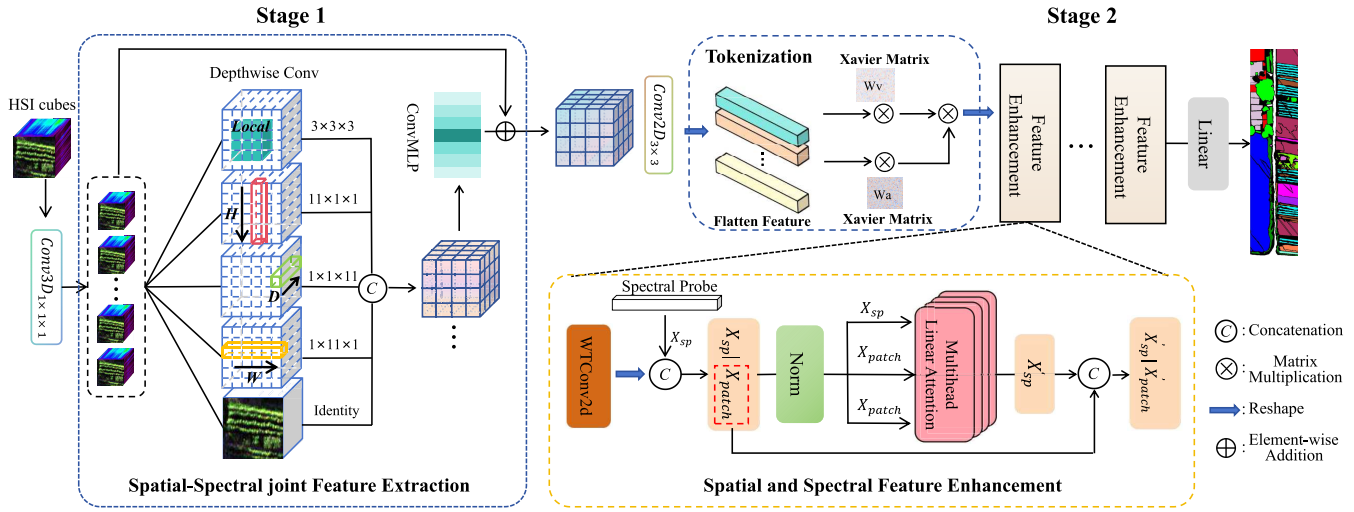


Fig. 2. Overall network framework diagram of the proposed DSW-LAN, including the DFDWConv3D block, the wavelet transform structure, and SLA layers.

that self-attention can be viewed as a special case of a cubic polynomial; they designed a polynomial operator that accesses the same data patterns as nonlocal blocks and self-attention modules while reducing complexity from quadratic to linear. Han et al. [46] proposed the Mamba-inspired linear attention model, incorporating Mamba's key designs—forget gating and block structuring—into a linear attention framework. Jiang et al. [47] proposed Mamba-like linear attention UNet, which combines linear attention with a Mamba-inspired adaptive mechanism to achieve linear computational complexity while maintaining high segmentation accuracy.

Building on these advancements, we proposed a linear attention mechanism, which retains the excellent key information reinforcement and redundant suppression capabilities of the attention mechanism while effectively reducing its complexity.

D. Methods Based on Mamba Mechanism

In recent years, the Mamba architecture based on SSM has demonstrated significant potential in HSI classification due to its linear complexity and robust long-range dependency modeling capabilities and offered an alternative to Transformers for HSI classification. For instance, Wang et al. [48] propose DBMGNet, which designs band selection enhanced bidirectional Mamba (BSEBM), leveraging Mamba's long-range dependence modeling and sequential modeling capabilities to process spatial–spectral information. Ahmad et al. [49] propose the morphological spatial Mamba and morphological spatial–spectral Mamba model, which combines the strengths of morphological operations and the SSM framework, offering a more computationally efficient alternative to Transformers. Zhang et al. [50] propose the Center-Scan Mamba network, which enables spatial–spectral long-range dependency modeling with linear complexity. Meanwhile, the center-scan mechanism is designed to enhance interactions between the central pixel and its surrounding pixels, effectively overcoming the limitations of traditional scanning methods. However, current Mamba-related research still has limitations in agricultural HSI scenarios, for example, its dynamic modeling capabilities for complex planting structures remain insufficiently validated.

III. Proposed Method

In this section, we provide a comprehensive explanation of the modules we designed. The overview of our proposed network model is illustrated in Fig. 2. In the spatial–spectral feature extraction stage, we designed DFDWConv3D to extract joint spectral–spatial features and capture long-range dependencies. During the spatial and spectral feature enhancement stage, we leverage invertible wavelet reconstruction to further refine spatial information in the frequency domain. Additionally, we proposed a 1-D spectral probe guided spectral attention mechanism with linear complexity, which amplifies key spectral information while maintaining computational efficiency.

A. Spatial–Spectral Feature Extraction

Three-dimensional CNNs have demonstrated powerful spectral–spatial modeling capabilities, particularly suited for context-aware tasks in complex agricultural scenarios, and are commonly used for primary feature extraction in HSI processing networks. However, conventional 3-D CNNs suffer from excessive parameters, high computational costs, and reliance on fixed kernel sizes, which restrict their receptive fields to local regions and easily neglect long-range dependencies. These limitations degrade the quality of spatial–spectral feature extraction, thereby compromising subsequent spatial and spectral feature enhancement and classification accuracy. To address these challenges, we proposed the DFDWConv3D block.

1) Factorized Kernel: We partition the 3-D input cube along the channel dimension into five groups, each processed by distinct convolutional kernels. A conventional $3 \times 3 \times 3$ kernel is employed in one group to extract localized spectral–spatial joint features. To enable large-context modeling while maintaining computational efficiency,

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we innovatively decompose the $11 \times 11 \times 11$ large kernel into three orthogonal 1-D components: a $1 \times 1 \times 11$ kernel along the spectral dimension captures cross-band dependencies, a $1 \times 11 \times 1$ kernel scans the spatial width direction to model horizontal patterns, and an $11 \times 1 \times 1$ kernel traverses the spatial height dimension to extract vertical structures. The fifth group retains the original features through an identity mapping pathway, serving as a spectral preservation mechanism to prevent distortion of agriculturally sensitive signatures.

First, assume that the input feature $X \in \mathbb{R}^{B\times C\times D\times H\times W}$, where B represents the batch size; C represents the number of channels; and D, H, and W corresponds the depth (spectral channels), height, and width of the input features, respectively. Divide X into groups along the channel dimension, denoted as g, represented as follows:

$$X_{\mathrm{HWD}}, X_{\mathrm{HW}}, X_{\mathrm{WD}}, X_{\mathrm{HD}}, X_{\mathrm{Id}} = \mathrm{Split_}X\,(g,\ g,\ g,\ g,\ C-4\times g) \tag{1}$$

where $g = C \times \mathrm{branch_ratio}$.

Then, we apply depthwise separable convolution with different kernel sizes to each group separately, which can be expressed as follows:

$$\begin{aligned} X'_{\mathrm{HWD}} &= \mathrm{DWConv3}\times 3\times 3\,(X_{\mathrm{HWD}}) \\ X'_{\mathrm{HW}} &= \mathrm{DWConv1}\times 1\times 11\,(X_{\mathrm{HW}}) \\ X'_{\mathrm{HD}} &= \mathrm{DWConv1}\times 11\times 1\,(X_{\mathrm{HD}}) \\ X'_{\mathrm{WD}} &= \mathrm{DWConv11}\times 1\times 1\,(X_{\mathrm{WD}}) \\ X'_{\mathrm{Id}} &= X_{\mathrm{Id}} \end{aligned} \tag{2}$$

where X'_{HWD} represents local features extracted by the cubic convolution kernel; X'_{HW} captures long-range cross-band dependencies through a depthwise elongated kernel; and X'_{HD} and X'_{WD} correspond to elongated kernels in the height and width directions, respectively, capturing broad spatial context. X'_{Id} remains unchanged, preserving the original identity without any operation.

Then, the outputs of all branches are concatenated along the channel dimension, which can be expressed as follows:

$$X_m = \mathrm{Concat}\left(X'_{\mathrm{HWD}}, X'_{\mathrm{HW}}, X'_{\mathrm{HD}}, X'_{\mathrm{WD}}, X'_{\mathrm{Id}}\right) \tag{3}$$

where X_m represents concatenated features.

2) Conv3DMLP: Depthwise separable convolution extracts spatial–spectral features only within channel groups, leading to information isolation between different groups. Therefore, an MLP is required for cross-channel interaction, integrating features extracted by different channel convolutions. However, applying a conventional MLP directly to a 3-D feature cube requires global pooling followed by a fully connected layer, which results in the loss of spatial information. To avoid this problem, we replace the fully connected layer in MLP with a $1 \times 1 \times 1$ 3-D convolution, significantly reducing both parameter count and computational cost. This approach avoids large intermediate vectors during flattening, preventing excessive memory consumption.

The detailed procedure of Conv3DMLP can be described as follows:

$$X_{\mathrm{hidden}} = \mathrm{GELU}\left(\mathrm{Norm}\left(\mathrm{Conv1}\times 1\times 1\,(X_m)\right)\right) \tag{4}$$

$$X'_m = \mathrm{Conv1}\times 1\times 1\left(\mathrm{Drop}\left(X_{\mathrm{hidden}}\right)\right) \tag{5}$$

where X_{hidden} is the hidden layer of MLP, Norm represents the layer normalization, GELU is the activation function, Drop refers to dropout, and X'_m refers to the output of Conv3DMLP.

Finally, a residual connection is applied to the initial features, producing the spatial–spectral feature output, which can be expressed as follows:

$$X_c = X'_m + X \tag{6}$$

where X_c is the final output of DFDWConv3D and + refers to element-wise addition.

B. Spatial and Spectral Feature Enhancement

Spatial–spectral features perform only a single joint extraction of spatial and spectral information, but they remain insufficient for fine-grained representation. Due to the lack of detailed characterization of local structures, texture patterns, and spectral variations, this extraction method struggles to distinguish crops with high similarity. In cases where spatial distribution structures are alike or spectral responses are close, spatial–spectral joint features often lack sufficient discriminative power, leading to reduced classification accuracy. Therefore, in the spatial and spectral feature enhancement stage, we separately refine spatial and spectral features.

1) Wavelet Transform: In agricultural scenarios, HSIs exhibit high class similarity and complex distributions, making it difficult to extract discriminative spatial features and limiting classification accuracy in boundary regions. Traditional methods face significant challenges in addressing these issues. To overcome this, we apply the wavelet transform to the feature maps; the Daubechies-4 (DB4) wavelet basis is selected for its optimal mathematical properties aligned with hyperspectral imaging (HSI) requirements. DB4's four vanishing moments ($\int x^k \psi(x)dx = 0,\ k = 0, 1, 2, 3$) enable precise capture of transient spatial features (e.g., crop boundaries) while preserving smooth spectral trends. Additionally, considering the multilevel wavelet transforms and inverse transforms in our network, the near symmetry of DB4 minimizes reconstruction distortion, which is critical for preserving spectral continuity in LL subbands and spatial details in high-frequency subbands.

The LL subband captures the overall structure and contours, while the LH, HL, and HH subbands retain fine-grained texture details. By separating these components, we enhance the distinction between similar crop features, mitigating challenges posed by overlapping frequency components in agricultural HSI. Additionally, wavelet transform functions exhibit directional sensitivity: LH, HL, and HH correspond to horizontal, vertical, and diagonal detail extraction, respectively. This improves responsiveness to crop planting patterns and spatial distribution, enabling more precise feature enhancement for precise classification.

Fig. 3 illustrates the detailed process of the wavelet transform convolution. Assume that the input tensor $F \in R^{c\times H\times W}$,

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and we perform a wavelet transform on it. The 2-D discrete wavelet transform can be mathematically expressed as follows:

$$\begin{cases} \mathrm{LL} = \phi(F) * \phi_{\mathrm{row}} * \phi_{\mathrm{col}} \\ \mathrm{LH} = \phi(F) * \phi_{\mathrm{row}} * \psi_{\mathrm{col}} \\ \mathrm{HL} = \phi(F) * \psi_{\mathrm{row}} * \phi_{\mathrm{col}} \\ \mathrm{HH} = \phi(F) * \psi_{\mathrm{row}} * \psi_{\mathrm{col}} \end{cases} \tag{7}$$

where $\phi(\cdot)$ is the low-pass filter and $\psi(\cdot)$ is the high-pass filter. If F undergoes n-level wavelet decomposition, the decomposition at level l is given by the following equation:

$$\mathrm{WT}^l(F) = \left[\mathrm{LL}^l, \mathrm{LH}^l, \mathrm{HL}^l, \mathrm{HH}^l\right] \in R^{C\times 4\times H_l\times W_l} \tag{8}$$

where LL^l denotes the LL subband at level l and $\mathrm{LH}^l, \mathrm{HL}^l$, and HH^l represent the high-frequency subbands. After decomposition, the tensor is downsampled according to $H_l = \lceil H_{l-1}/2 \rceil$ and $W_l = \lceil W_{l-1}/2 \rceil$.

Then, the four subbands are merged into a $c \times 4$ channel tensor and processed independently using depthwise separable convolution with groups $= c \times 4$, as in the following equation:

$$Y_{\mathrm{sub}}^{(i)} = \mathrm{DWConv5} \times 5\left(\mathrm{WT}^l(F)\right) \odot \alpha^{(i)} \tag{9}$$

where $\alpha^{(i)} \in R^{c\times 4}$ is a learnable scaling parameter, allowing the network to automatically learn the contribution weight of each subband. Additionally, $i \in \{1, 2, 3, 4\}$ represents the subband index.

After that, the LL subband LL^l undergoes the next level of decomposition, forming a pyramid structure, which can be expressed as follows:

$$\mathrm{WT}^{l+1}\left(\mathrm{LL}^l\right) = \left[\mathrm{LL}^{l+1}, \mathrm{LH}^{l+1}, \mathrm{HL}^{l+1}, \mathrm{HH}^{l+1}\right]. \tag{10}$$

When reaching the deepest level, the inverse transform is applied to progressively reconstruct the LL components as follows:

$$\mathrm{LL}_{\mathrm{rec}}^l = \mathrm{IWT}\left(\left[\mathrm{LL}^l, \mathrm{LH}^l, \mathrm{HL}^l, \mathrm{HH}^l\right]\right) + \mathrm{LL}_{\mathrm{rec}}^{l+1}. \tag{11}$$

The final output of wavelet domain is $F_{\mathrm{rec}} = \mathrm{LL}_{\mathrm{rec}}^0$.

The final output of the wavelet transform convolution module is a linear combination of spatial-domain base convolution and wavelet-domain features, which can be expressed as follows:

$$X_{\mathrm{patch}} = \mathcal{C}_{\mathrm{base}}(F) + F_{\mathrm{rec}} \tag{12}$$

where $\mathcal{C}_{\mathrm{base}}(F)$ represents the spatial-domain features extracted through base convolution, F_{rec} represents the reconstructed wavelet-domain features, and $X_{\mathrm{patch}} \in R^{c\times H\times W}$ refers to the output of wavelet transform convolution.

In summary, the multilevel decomposition and reconstruction of wavelet transform enable layer-wise extraction and fusion of complex spatial information in HSIs. This allows the model to capture key discriminative features across different scales and levels, resulting in a more robust and highly distinctive feature representation.

Algorithm 1 Dual-Stage Wavelet Enhancement and Linear Attention Network for HSI Classification

Input: The original HSIs.

Output: Predicted labels of all categories.

Initialization:

batch size is 64; loss function is Cross-Entropy; number of channels after PCA is 30; patch size is 13; optimizer is Adam; learning rate is $1e-3$.

Repeat:

1: Perform PCA reducing dimension to 30;
2: Perform a $1 \times 1 \times 1$ convolution layer to adjust channel number;
3: Perform the DFDWConv3D and the ConvMLP to get spatial-spectral feature;
4: Perform a 3×3 convolution layer;
5: Flatten features into 1D vector and perform tokenizer block to get spatial-spectral tokens;
6: Reshape the tokens and perform the wavelet tranform block;
7: Concatenate a 1D vector Q and perform the linear attention layer to get classification token;
8: Perform the Cat operation to get final fused feature;
9: Perform the linear layer to get labels;

end for: Get the classification result of testing samples.

2) Spectral Linear Attention: In complex agricultural scenarios, feature confusion caused by spectral overlap and high interclass similarity is particularly prominent, typically manifested as reflectance convergence of different crops in visible bands, necessitating more effective methods for spectral feature extraction. Notably, materials of the same category may exhibit stronger responses in certain spectral bands while overlapping with other categories in other bands. Furthermore, significant long-range dependencies across distant bands also exist, exemplified by the statistical correlation between the 680-nm reflectance and the 1450-nm reflectance in agricultural remote sensing. Moreover, discriminative features are sparsely distributed in specific narrowbands, such as crop stress responses concentrated in partial red-edge regions, resulting in the challenge of dynamically sparse discriminative spectral features. Conventional attention methods struggle to balance global modeling and computational efficiency for large-scale applications.

Therefore, we propose probe-guided spectral linear attention (SLA) as is shown in Fig. 4. To address the dynamically sparse discriminative spectral features and spectral overlap and high interclass similarity in complex agricultural scenes, our spectral probe extracts full-band information via learnable low-dimensional vector and dynamically amplifies class-specific responses while suppressing irrelevant bands, explicitly enhancing interclass discriminability, concentrating attention resources on critical bands. To resolve long-range dependencies across spectrally distant bands, SLA's kernel attention directly models interband correlations beyond wavelength proximity, capturing agriculturally driven nonlocal interactions. Besides, we use a 1-D spectral probe to interact with the original features, rather than performing self-

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TABLE I
DETAILED OUTPUT SIZE OF ALL STEPS ON THE WHU-HI-HANCHUAN DATASET

Step No.	Step Name	Output Size
1	Original HSI	$1217 \times 303 \times 274$
2	Reduce Dimension (PCA=30)	$1217 \times 303 \times 30$
3	Extract Patch	$13 \times 13 \times 30$
4	Perform $3D$ convolution	$8\ 13 \times 13 \times 30$
5	Perform DFDWConv3D block	$8\ 13 \times 13 \times 30$
6	Reshape	$240\ 13 \times 13$
7	Perform $2D$ convolution	$64\ 11 \times 11$
8.1	Flatten	64 121
8.2	Perform Tokenizer	32 64
9.1	Reshape and perform wavelet block	$32\ 8 \times 8$
9.2	Perform linear attention	$T \in \boldsymbol{R}^{4\times 64}$
10	Perform Cat	33 64
11	Perform linear layer	$Lable \in \boldsymbol{R}^{16\times 64}$

attention calculations entirely on the raw features, reducing the computational complexity of attention from $O(N^2)$ to $O(N)$, achieving an optimal balance between classification accuracy and computational efficiency, enabling real-time spectral feature enhancement.

First, we initialize a 1-D vector as the spectral probe $Q \in R^{1\times c}$ and concatenate it with the flattened feature Y; the linear weight projections for keys and values are formulated as follows:

$$Y_c = \text{Concat}\,(Q, \text{Flatten}\,(Y)) \in R^{(N+1)\times c} \tag{13}$$

$$K = YW_K, V = YW_V \in R^{N\times c} \tag{14}$$

where Y_c is the concatenated feature Y and $W_K, W_V \in R^{c\times c}$ are the linear projection matrices for the key and value, respectively.

Next, we perform multihead decomposition by splitting Q, K, and V into h attention heads as follows:

$$\begin{aligned} Q_h &= \text{Reshape}\,(Q) \in R^{1\times d} \\ K_h &= \text{Reshape}\,(K) \in R^{N\times d} \\ V_h &= \text{Reshape}\,(V) \in R^{N\times d} \end{aligned} \tag{15}$$

where $d = C/h$ represents the head dimension.

Subsequently, linear attention is applied to model global spectral relationships. The attention is computed as follows:

$$\begin{aligned} A &= \varepsilon \bullet \left(Q_h K_h^T\right) \in R^{h\times 1\times N} \\ \text{Score} &= \text{Softmax}\,(A, \text{dim} = -1) \\ \mathcal{Z}_h &= \text{Score} \bullet V_h \in R^{h\times 1\times d} \end{aligned} \tag{16}$$

where $\varepsilon = \sqrt{d}$ is the scaling factor and $\mathcal{Z}_h$ represents the output of head h. After merging the multihead attention outputs and reshaping the features, the final output of the attention mechanism can be described as follows:

$$X_{\text{sp}} = \text{Concat}\,(\mathcal{Z}_1, \ldots, \mathcal{Z}_h) \in R^{1\times c} \tag{17}$$

where X_{sp} is the final output of SLA.

Finally, by stacking the mentioned spatial and spectral feature enhancement modules, we can progressively extract and fuse complex spatial–spectral information from HSIs. As we can see in Fig. 2, the output of each spatial and spectral feature enhancement module is formed by concatenating the output of WTConv X_{patch} (which contains refined spatial features) and the output of SLA X_{sp} (which contains refined spectral features), and these components subsequently undergo further

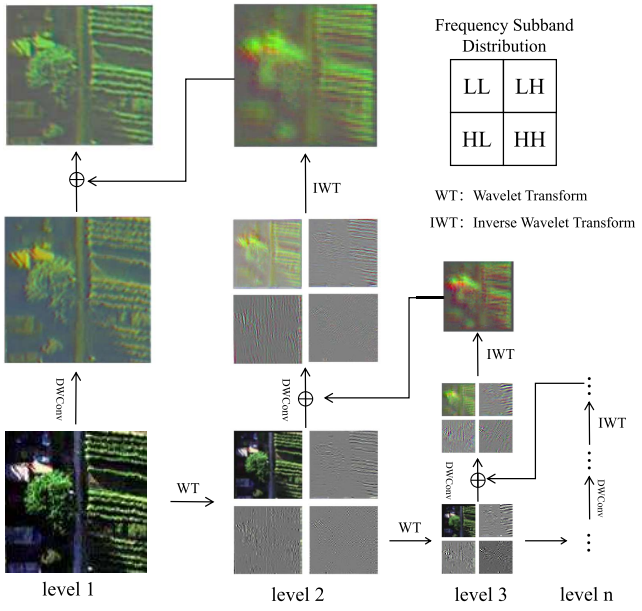


Fig. 3. Structural details of the wavelet transform convolution module. This module integrates wavelet transform with depthwise separable convolution, performing multilevel decomposition of the image in the wavelet domain, processing it through convolution layers, and then reconstructing the image to extract more discriminative spatial features.

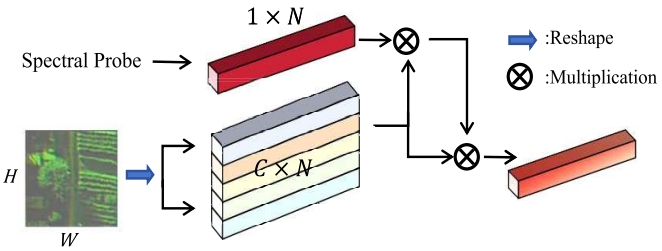


Fig. 4. Structural details of the SLA module. This module primarily implements the spectral probe's attention weight allocation across all bands, emphasizing the most important ones while capturing long-range spectral dependencies.

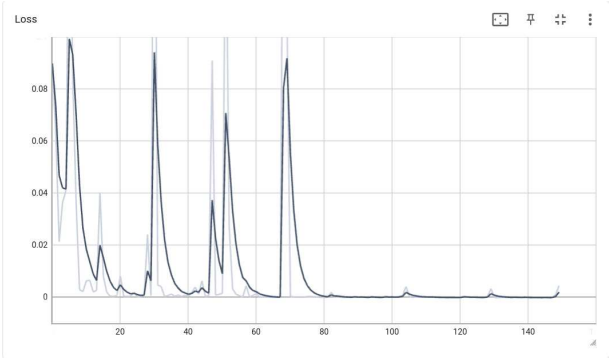


Fig. 5. Loss curve at 150 training epochs.

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TABLE II
DETAILS AND PROPORTION OF SAMPLES DIVISION OF THREE DATASETS

Dataset Name	Spatial Resolution	Bands Number	Classes Number	Pixel Size	Training Proportion	Testing Proportion
Indian Pines	20m	200	16	145×145	5%	95%
WHU-Hi-HongHu	0.043m	270	22	940×475	1%	99%
WHU-Hi-HanChuan	0.109m	274	16	1217×303	1%	99%

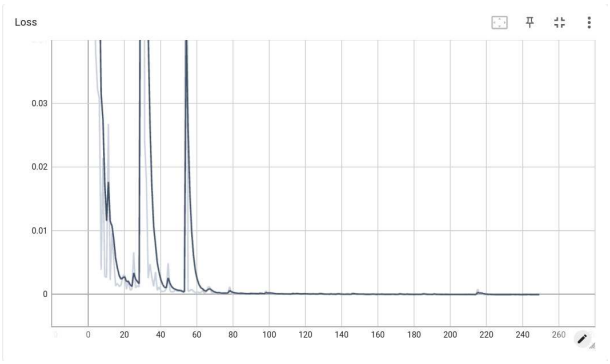


Fig. 6. Loss curve at 250 training epochs.

feature enhancement separately in the next layer. Each layer of the spatial and spectral feature enhancement module enables deep interaction between local and global features, maximizing the extraction of discriminative spatial–spectral characteristics.

C. Collaborative Interactions

In summary, the interaction between the three main components of our model creates a cohesive framework that enhances the classification accuracy of HSIs in complex agricultural scenarios. The DFDWConv3D module extracts spatial–spectral joint features and captures local patterns using both conventional and decomposed convolutional kernels, providing essential spatial information for differentiating similar crop types.

The wavelet transform convolution module further enhances spatial features by decomposing feature maps into various wavelet subbands, effectively separating coarse structures from fine details, which improves the discriminability of similar crop features and addresses interclass similarity challenges.

Finally, the SLA dynamically amplifies significant spectral bands while suppressing irrelevant ones, ensuring the model focuses on the most discriminative features. SLA also models long-range dependencies between spectrally distant bands, capturing vital nonlocal interactions driven by agricultural features.

The synergy among these components fosters a deeper understanding of complex agricultural data relationships, significantly boosting the model's classification capabilities while maintaining a balance between computational efficiency and accuracy, thus making it suitable for real-time applications in agriculture.

D. Implementation

To enhance the understanding of the network architecture, we briefly outline the model's execution steps in Algorithm 1 and provide the detailed output dimensions for each stage in Table I.

TABLE III
COMPARATIVE RESULTS OF DIFFERENT KERNEL SIZES IN DFDWCONV3D ON WHU-HI-HANCHUAN. THE BEST RESULTS ARE OUTLINED IN BOLD

Metric	The size of different kernel			
	7	9	11	13
OA (%)	97.57±0.24	97.55±0.15	**97.72±0.16**	97.50±0.53
AA (%)	94.80±0.26	94.84±0.34	**95.27±0.53**	94.72±1.16
κ*100	97.16±0.29	97.13±0.18	**97.34±0.19**	97.07±0.63

TABLE IV
COMPARATIVE RESULTS OF LEVEL IN WAVELET TRANSFORM ON WHU-HI-HANCHUAN. THE BEST RESULTS ARE OUTLINED IN BOLD

Metric	The level of wavelet transform			
	1	2	3	4
OA (%)	97.25±0.69	97.37±0.15	**97.72±0.16**	97.50±0.36
AA (%)	94.03±1.40	94.53±0.31	**95.27±0.53**	94.82±0.62
κ*100	96.79±0.81	96.92±0.18	**97.34±0.19**	97.08±0.42
Parameters(M)	196704	203360	210016	216672
FPS(frames/sec)	248.96	193.36	179.33	153.69

TABLE V
COMPARATIVE RESULTS OF NUMBER OF SPATIAL AND SPECTRAL FEATURE ENHANCEMENT BLOCKS ON WHU-HI-HANCHUAN. THE BEST RESULTS ARE OUTLINED IN BOLD

Metric	The number of spatial and spectral feature enhancement			
	1	2	3	4
OA (%)	97.18±0.54	**97.72±0.16**	97.61±0.34	97.35±0.43
AA (%)	94.37±0.96	**95.27±0.53**	94.83±0.77	94.82±0.62
κ*100	96.70±0.64	**97.34±0.19**	97.21±0.40	97.08±0.42
Parameters(M)	182464	210016	237568	265120
FLOPs(G)	1.47	1.50	1.53	1.55
FPS(frames/sec)	270.49	193.36	132.95	102.83

IV. EXPERIMENTS

In this section, we will introduce three HSI classification datasets used in our experiments. Based on these datasets, we conducted comprehensive comparative experiments and ablation studies to validate the effectiveness and generalization capability of our network. To mitigate experimental randomness, all reported results were obtained by averaging ten independent trials, with standard deviations calculated to quantify statistical significance.

A. HSI Datasets

We conducted experiments on three public hyperspectral datasets: WHU-Hi-HanChuan, WHU-Hi-HongHu, and Indian

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TABLE VI

COMPARISON OF DSW-LAN USING DIFFERENT ATTENTION AND DFDWCONV3D ON WHU-HI-HANCHUAN. THE BEST RESULTS ARE OUTLINED IN BOLD

Metric	CBAM	SimAM	EMA	CCA	LiteMLA	CSCA	DFDWCONV3D
OA (%)	97.01±0.49	97.27±0.17	97.15±0.26	97.50±0.39	97.34±0.43	97.16±0.44	**97.72±0.16**
AA (%)	93.43±1.32	94.00±0.56	93.94±0.66	94.45±1.35	94.43±0.89	94.16±0.63	**95.27±0.53**
κ*100	96.50±0.58	96.81±0.20	96.66±0.30	97.08±0.46	96.89±0.51	96.67±0.52	**97.34±0.19**

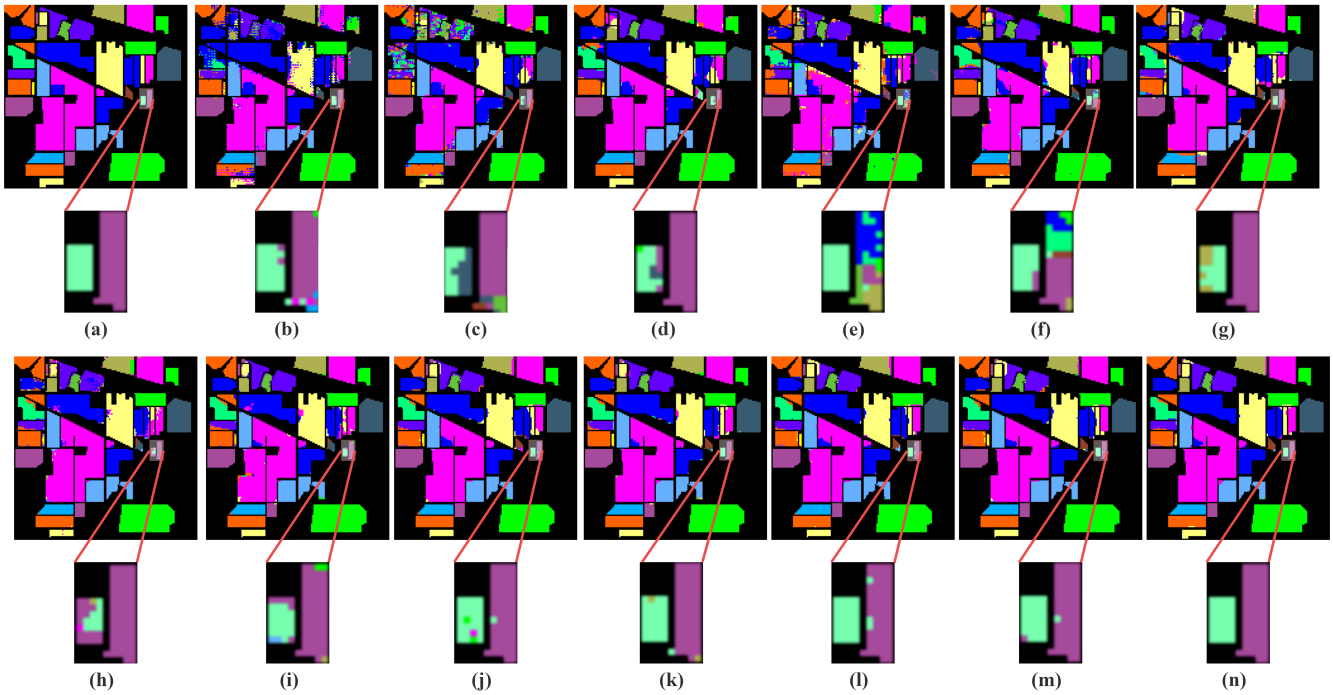


Fig. 7. Visualization of different DL-based networks on Indian Pines. (a) Ground truth. (b) Resnet [51]. (c) ContextualNet [52]. (d) SSRN [53]. (e) SSFTT [54]. (f) LSFAT [55]. (g) MATNet [56]. (h) SANet [57]. (i) SS-MSDCNet [58]. (j) MSFI-CNet [59]. (k) IGroupSS-Mamba [60]. (l) DBMGNet [48]. (m) CSCANet [44]. (n) Proposed DSW-LAN.

TABLE VII

COMPARISON OF DSW-LAN USING DIFFERENT TRANSFORMERS AND SLA ON WHU-HI-HANCHUAN. THE BEST RESULTS ARE OUTLINED IN BOLD

Metric	morphFormer	SPTNet	DSFormer	MamTrans	SLA
OA (%)	97.07±0.30	97.47±0.11	97.14±0.43	97.43±0.39	**97.72±0.24**
AA (%)	93.42±1.38	94.52±0.18	94.20±0.84	94.66±0.43	**95.27±0.53**
κ*100	96.57±0.35	97.04±0.13	96.66±0.51	96.99±0.28	**97.34±0.19**

TABLE VIII

ABLATION RESULTS OF DIFFERENT MODULES ON WHU-HI-HANCHUAN. THE BEST RESULTS ARE OUTLINED IN BOLD

Metric	WHU-Hi-HanChuan			
DFDWConv		✓	✓	✓
WT	✓		✓	✓
SLA	✓	✓		✓
OA (%)	97.18±0.72	96.84±1.37	96.55±1.16	**97.72±0.16**
AA (%)	94.10±1.50	93.39±2.40	92.95±1.90	**95.27±0.53**
κ*100	96.70±0.84	96.30±1.60	95.96±1.35	**97.34±0.19**
FLOPs(G)	1.19	1.52	1.48	1.55
FPS(frames/s)	109.40	167.63	82.04	77.15
Size(MB)	85.62	284.10	290.04	303.67

Pines. Among these, the WHU-Hi-HanChuan and WHU-Hi-HongHu datasets were acquired from agricultural scenes characterized by rich category distributions and complex spatial structures, which pose significant classification challenges. Additionally, we selected the widely used dataset Indian Pines to validate the generalization capability of our proposed method on standard public datasets. Details of these datasets, including training–test split ratios, are provided in Table II.

TABLE IX

ABLATION RESULTS OF DIFFERENT MODULES ON WHU-HI-HONGHU. THE BEST RESULTS ARE OUTLINED IN BOLD

Metric	WHU-Hi-HongHu			
DFDWConv		✓	✓	✓
WT	✓		✓	✓
SLA	✓	✓		✓
OA (%)	98.13±0.30	98.04±0.65	98.13±0.12	**98.37±0.14**
AA (%)	95.87±0.87	95.62±1.11	95.82±0.41	**96.40±0.28**
κ*100	97.63±0.37	97.53±0.81	97.63±0.15	**97.94±0.18**

B. Experimental Setup

1) Evaluation Metrics: The three evaluation indicators we used are common in the HSI classification task, which are

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TABLE X

DETAILED CLASSIFICATION ACCURACY OF INDIAN PINES ON DIFFERENT DL-BASED NETWORKS. THE BEST RESULTS ARE OUTLINED IN BOLD

Class No.	ResNet	ContextualNet	SSRN	SSFTT	LSFAT	MATNet	SANet	SS-MSDCNet	MSFI-CNet	IGroupSS-Mamba	DBMGNet	CSCANet	DSW-LAN
1	94.05±12.97	72.97±18.54	38.75±47.58	88.10±18.77	90.57±6.86	81.16±24.25	12.55±4.67	79.53±16.99	62.95±10.16	50.70±12.85	62.72±13.02	80.69±6.90	63.18±16.16
2	77.58±12.86	91.24±5.28	96.38±1.69	90.52±3.43	93.74±2.16	92.79±1.27	90.87±3.11	96.04±0.82	94.50±0.49	94.92±0.81	96.33±0.71	95.01±0.92	**96.39±0.84**
3	83.10±11.85	73.97±8.98	93.79±2.63	86.98±4.08	92.75±4.48	95.64±1.27	97.98±1.43	94.55±2.94	97.73±0.90	**99.15±0.59**	99.06±0.49	97.41±0.85	97.81±0.67
4	91.64±13.53	66.54±14.21	94.23±3.59	90.08±7.17	91.80±5.63	91.88±2.92	79.78±8.72	93.66±4.37	92.93±2.06	91.29±3.65	**94.75±2.41**	92.99±1.37	**93.91±1.50**
5	97.05±2.44	88.34±9.76	97.73±1.15	91.52±4.37	97.59±2.17	98.71±2.10	98.69±0.75	98.74±0.94	**99.32±0.64**	98.73±0.75	**99.82±0.21**	98.75±0.66	98.93±0.60
6	91.08±5.16	94.57±3.13	96.87±1.79	95.24±2.22	98.17±1.28	98.63±0.56	**99.31±0.32**	96.73±1.07	99.00±0.30	98.22±0.48	98.96±0.49	98.69±0.39	98.31±0.50
7	76.74±20.22	74.95±11.15	68.11±38.78	64.04±19.37	76.45±13.73	91.85±12.26	6.30±10.08	72.96±29.49	65.92±16.29	92.96±7.30	92.96±9.72	88.88±13.35	**98.15±3.41**
8	95.17±2.98	94.10±4.64	97.25±3.94	97.78±1.20	97.90±1.88	99.78±0.22	**99.96±0.09**	99.93±0.14	99.73±0.21	99.93±0.10	98.92± 1.00	99.93±0.19	99.71±0.33
9	54.10±36.83	46.93±14.81	77.45±30.09	45.56±25.55	85.61±11.87	69.47±13.06	22.11±17.10	86.84±11.83	**87.36±4.82**	87.36±6.74	78.94±11.28	85.26±16.44	79.47±14.60
10	89.86±6.01	89.19±3.64	93.18±2.91	88.92±3.30	88.34±6.01	95.62±1.42	91.92±3.65	92.26±4.87	96.99±0.58	96.52±0.77	97.78±0.77	97.53±0.49	**98.03±0.70**
11	82.12±7.99	91.74±3.50	97.20±1.09	94.56±1.62	96.13±1.20	97.52±1.16	97.94±1.05	98.22±0.59	**98.94±0.37**	98.33±0.40	98.82±0.17	98.84±0.34	98.60±0.47
12	87.69±8.24	66.04±8.98	94.58±3.11	82.84±5.41	90.10±3.78	86.98±1.65	83.07±8.73	88.56±3.24	89.98±1.30	93.03±1.56	91.61±2.09	**93.05±1.43**	91.44±1.42
13	93.02±5.17	76.59±8.55	98.31±1.81	93.75±8.13	96.78±3.91	97.64±0.52	98.97±1.26	99.69±0.47	96.05±2.31	99.58±0.44	97.53±2.29	98.41±0.53	**99.95±0.15**
14	91.79±3.72	93.66±3.98	97.55±1.79	96.32±2.47	97.18±2.11	98.91±0.53	**99.91±0.16**	99.40±0.53	99.63±0.21	99.53±0.27	99.40±0.12	99.38±0.31	99.40±0.37
15	92.32±5.06	78.13±10.43	95.52±2.85	92.29±3.77	94.17±2.67	93.46±3.59	86.81±4.70	89.59±4.45	94.85±1.79	93.65±2.88	94.16±2.11	95.53±1.82	**97.55±1.32**
16	**95.04±5.63**	72.26±17.35	89.84±13.28	79.14±9.30	79.27±9.30	82.99±10.17	81.72±9.65	94.37±5.02	75.90±3.08	87.70±3.60	81.25±7.29	86.55±8.49	82.84±2.98
OA (%)	84.55±2.40	87.24±2.23	96.00±0.92	91.53±1.23	94.22±0.88	95.77±0.88	94.25±0.98	96.03±0.82	96.83±0.22	97.02±0.30	97.44±0.28	97.32±0.24	**97.44±0.24**
AA (%)	87.02±4.04	79.45±2.75	89.17±5.65	86.10±2.58	91.66± 1.76	92.06±3.24	77.99±2.53	92.57±2.51	90.74±1.09	92.60±0.79	92.69±1.53	**94.18±1.30**	93.35±1.68
κ*100	82.18±2.84	85.41±2.57	95.43±1.05	90.35±1.40	93.41± 0.10	95.18±1.00	93.43±1.13	95.46±0.94	96.38±0.25	96.60±0.34	97.07±0.32	96.94±0.27	**97.08±0.27**

TABLE XI

DETAILED CLASSIFICATION ACCURACY OF WHU-HI-HONGHU ON DIFFERENT DL-BASED NETWORKS. THE BEST RESULTS ARE OUTLINED IN BOLD

Class No.	ResNet	ContextualNet	SSRN	SSFTT	LSFAT	MATNet	SANet	SS-MSDCNet	MSFI-CNet	IGroupSS-Mamba	DBMGNet	CSCANet	DSW-LAN
1	96.86±2.74	96.42±3.33	98.31±0.97	96.93±0.92	97.99±1.10	97.37±0.49	98.14±0.38	**98.45±0.37**	97.94±0.65	98.16±0.39	97.75±1.65	98.42±0.15	98.06±0.31
2	73.79±9.82	76.49±8.03	84.56±4.57	82.22±5.42	86.11±5.18	83.78±5.38	85.50±3.31	**91.57±1.91**	88.41±1.86	92.07±1.11	84.57±15.25	90.04±2.05	87.96±2.28
3	85.68±6.50	87.80±2.97	91.33±2.88	96.40±0.59	96.61±0.88	95.95±1.39	97.35±0.89	**98.57±0.39**	98.23±0.40	98.24±0.43	97.99±0.95	97.86±0.41	97.61±0.34
4	96.41±1.75	97.96±0.78	98.51±0.69	99.67±0.31	99.70±0.14	99.69±0.24	99.89±0.11	99.90±0.05	**99.92±0.02**	99.81±0.05	99.84±0.14	99.91±0.02	99.90±0.05
5	77.14±21.46	88.20±4.01	87.95±7.03	91.30±2.05	92.25±2.40	92.06±4.91	84.39±3.87	95.82±2.62	95.07±1.63	93.79±1.82	98.90±0.79	94.20±1.39	**96.28±1.15**
6	95.55±3.89	96.67±1.12	97.20±1.09	99.23±0.34	99.31±0.38	99.20±0.35	99.29±0.52	98.48±0.91	99.37±0.14	99.41±0.25	99.61± 0.12	**99.59±0.06**	99.52±0.12
7	78.76±11.59	84.51±4.11	89.81±3.76	94.84±0.72	95.11±1.45	94.83±1.93	97.59±0.77	95.51±1.13	97.55±0.78	96.22±0.25	97.43±1.13	97.74±0.36	**97.89±0.43**
8	57.74±18.36	56.62±12.30	67.82±10.66	83.54±4.79	86.23±1.86	86.08±4.53	85.57±5.37	82.21±4.62	91.16±2.06	89.96±3.08	92.57±3.76	91.33±1.27	**93.53±0.66**
9	97.40±2.75	96.71±2.42	98.84±0.63	97.76±0.93	98.93±0.90	98.42±0.62	98.85±0.60	99.13±0.43	99.01±0.43	99.43±0.40	99.45±0.29	**99.27±0.43**	99.03±0.19
10	85.93±16.86	87.01±5.39	92.37±4.98	93.92±1.34	91.38±4.29	93.35±1.93	95.51±1.42	93.09±5.30	95.33±0.90	95.48±0.91	97.20±0.65	**97.30±0.53**	96.76±0.46
11	77.08±11.90	82.77±7.65	82.67±5.79	91.61±1.81	89.06±4.14	91.57±4.79	93.66±1.35	86.39±5.95	95.67±1.19	95.16±0.86	95.11±3.00	95.01±0.77	**96.05±0.97**
12	61.12±13.32	78.68±5.58	78.42±9.42	90.04±2.62	88.77±3.21	87.36±4.35	89.11±4.08	92.92±4.01	91.48±1.61	92.78±1.43	96.10±1.90	94.73±0.93	**95.24±2.29**
13	75.46±12.00	80.46±5.16	88.34±4.36	91.18±2.15	91.29±1.91	92.79±2.03	93.47±1.68	92.82±1.97	94.81±0.43	93.75±0.94	95.22±1.06	95.23±0.26	**95.45±0.99**
14	87.79±10.36	88.89±5.84	90.05±5.82	93.09±1.75	93.61±4.03	91.90±4.31	94.19±1.53	94.03±3.92	96.70±0.89	97.02±0.98	97.65±2.00	97.09±0.68	**97.73±0.54**
15	92.48±17.76	84.28±14.90	93.23±12.87	83.80±6.61	87.65±4.88	85.70±9.36	80.65±5.22	89.47±2.51	91.97±3.73	90.60±3.03	95.13±1.23	89.29±1.73	**94.74±1.35**
16	96.38±2.57	93.46±5.02	95.47±8.30	97.75±1.64	97.67±1.01	97.99±1.54	98.83±1.05	98.44±1.68	98.83±0.56	98.78±0.52	99.20±0.33	**99.20±0.29**	98.56±0.75
17	79.94±7.98	80.11±8.35	80.82±14.82	97.48±1.36	93.50±5.95	96.28±1.81	97.89±1.58	**99.27±0.75**	98.41±0.74	96.32±1.62	98.39±0.91	97.66±3.97	99.08±0.59
18	83.76±14.20	84.06±10.33	84.41±6.69	87.20±5.79	92.70±4.47	89.26±5.37	95.20±2.42	94.46±1.19	95.85±1.28	95.97±1.02	88.08±16.85	**96.09±1.30**	94.63±1.50
19	84.97±9.57	88.33±6.92	92.19±4.91	92.66±2.56	91.79±4.17	93.02±1.31	93.11±1.24	**98.15±0.61**	94.83±0.94	94.54±1.15	95.95±0.91	96.93±0.85	94.69±0.67
20	57.27±17.64	75.31±10.49	89.51±5.87	85.57±3.88	88.58±4.09	86.44±3.95	89.63±3.72	92.81±2.65	92.95±2.34	95.22±1.66	91.11±10.69	93.10±0.78	**93.63±1.89**
21	27.04±13.92	31.06±8.18	24.82±19.70	93.09±3.38	82.93±8.65	85.19±9.57	86.69±6.94	75.00±12.54	93.31±2.14	91.73±3.93	92.86±1.84	93.22±4.67	**95.70±2.10**
22	86.90±12.19	85.23±3.90	86.62±6.97	91.48±3.17	94.45±4.82	92.74±4.07	95.23±3.18	98.44±0.66	95.27±1.47	98.91±0.49	98.21±2.39	98.24±0.85	**98.66±0.98**
OA (%)	87.51±2.74	92.05±0.44	93.98±0.70	96.67±0.23	96.73±0.50	96.75±0.62	97.40±0.32	97.39±0.46	98.09±0.19	97.97±0.10	98.29±0.63	98.35±0.11	**98.37±0.14**
AA (%)	79.79±3.22	82.77±1.49	86.06±2.73	92.30±0.63	92.53±1.39	92.32±1.44	93.17±1.23	93.86±1.08	95.55±0.38	95.61±0.17	95.83±1.82	95.98±0.42	**96.40±0.28**
κ*100	84.17±3.33	89.91±0.58	92.37±0.89	95.79±0.29	95.87±0.63	95.89±0.78	96.71±0.40	96.70±0.59	97.58±0.24	97.44±0.12	97.83±0.80	97.91±0.13	**97.94±0.18**
Params	21380630	554128	364168	426945	782724	1708400	1086370	19451769	401052	**139982**	1535670	824161	210016
FLOPs(G)	3.14	2.02	10.14	0.73	**0.54**	2.81	2.01	168.47	2.12	0.61	11.10	1.63	1.55
FPS(frames/sec)	35.88	8.65	3.15	**657.77**	169.78	59.28	17.19	7.23	33.44	3.06	64.65	227.63	79.63
Total Size(MB)	84.12	**3.28**	23.40	36.14	49.66	218.62	203.65	4928.78	221.90	188.87	672.66	93.51	303.67

TABLE XII

DETAILED CLASSIFICATION ACCURACY OF WHU-HI-HANCHUAN ON DIFFERENT DL-BASED NETWORKS. THE BEST RESULTS ARE OUTLINED IN BOLD

Class No.	ResNet	ContextualNet	SSRN	SSFTT	LSFAT	MATNet	SANet	SS-MSDCNet	MSFI-CNet	IGroupSS-Mamba	DBMGNet	CSCANet	DSW-LAN
1	93.09±2.62	93.51±1.66	93.70±2.49	93.66±1.65	95.97±1.43	96.85±0.54	94.95±1.96	98.77±0.64	99.35±0.20	99.08±0.34	99.32±0.35	99.30±0.10	**99.39±0.11**
2	94.66±2.44	91.05±3.33	91.16±5.01	93.90±2.54	93.88±3.01	94.74±2.90	95.46±1.37	95.71±1.80	94.85±1.75	96.12±0.72	94.11±2.07	96.31±0.95	**96.70±0.62**
3	84.31±8.58	85.12±4.44	88.37±6.33	93.33±2.53	90.31±7.26	95.86±2.42	95.67±2.00	**98.46±1.35**	96.64±1.34	96.29±0.96	96.47±4.13	97.43±0.64	98.41±0.55
4	92.01±6.58	93.57±5.48	94.36±6.98	98.25±1.65	96.24±2.63	97.92±1.10	98.53±0.62	**99.53±0.51**	99.01±0.31	98.56±0.31	99.33±0.23	99.35±0.25	99.23±0.15
5	80.94±12.03	62.56±18.94	41.08±36.02	81.58±9.26	90.47±6.85	76.82±16.78	88.48±8.35	**95.98±1.66**	90.34±3.58	88.10±4.31	93.67±2.71	95.10±1.20	94.33±2.50
6	77.29±7.88	61.92±9.39	64.29±11.57	78.47±6.72	84.27±6.74	80.11±9.08	**87.67±2.75**	82.75±7.00	82.28±3.57	83.49±4.49	87.29±4.91	80.38±1.59	86.53±1.20
7	78.61±10.49	81.82±9.75	82.02±9.24	84.52±7.85	90.83±5.37	88.62±9.05	89.45±5.11	93.33±2.65	90.62±4.42	92.48±0.46	92.99±1.01	**94.43±1.20**	93.95±0.75
8	85.79±6.13	84.16±3.96	85.99±4.31	90.48±2.99	92.19±5.46	93.27±3.75	94.54±1.09	94.63±2.75	93.80±3.95	96.21±0.91	96.32±0.48	95.46±0.99	**96.46±0.60**
9	77.49±5.48	77.68±8.08	82.80±5.57	84.96±6.83	88.79±2.94	88.75±7.40	89.92±2.88	96.02±1.72	94.87±2.52	96.75±1.48	97.05±0.84	96.80±0.55	**97.13±0.97**
10	97.91±2.09	96.69±1.90	98.23±1.21	98.04±0.98	97.23±3.00	98.62±0.87	97.79±2.00	98.40±0.60	98.47±0.60	99.17±0.35	**99.63±0.23**	98.55±0.29	98.76±0.43
11	85.50±7.26	92.23±4.24	93.25±2.42	94.19±2.14	96.54±1.60	93.99±3.04	95.51±1.54	95.62±4.36	97.95±0.41	98.47±0.54	97.94±0.79	**99.09±0.23**	98.40±0.33
12	69.63±16.63	59.89±8.07	43.83±23.68	81.87±5.11	84.48±8.94	85.14±5.75	86.89±8.09	85.95±9.71	89.58±2.74	**94.67±1.69**	91.89±5.47	94.51±1.40	93.98±2.06
13	73.83±10.05	72.15±12.51	74.63±11.45	84.06±4.45	78.97±8.79	86.61±4.83	**89.07±4.15**	78.11±13.64	80.66±2.39	84.26±2.40	84.34±4.18	84.99±1.34	86.33±2.08
14	84.63±5.82	87.17±1.93	87.44±5.96	93.55±2.73	93.02±2.53	95.55±3.50	94.65±1.75	**97.29±1.20**	94.63±2.68	95.90±0.54	96.23±2.81	96.49±0.36	97.13±0.48
15	83.02±10.75	79.37±11.56	67.95±39.47	88.63±14.55	88.35±8.00	78.84±17.47	**93.41±6.78**	64.36±6.12	88.79±3.71	75.84±2.32	83.81±5.13	84.53±3.37	87.80±6.57
16	96.75±4.08	99.34±0.50	99.86±0.10	99.77±0.07	99.61±0.20	99.51±0.57	99.76±0.13	99.88±0.07	99.81±0.07	**99.89±0.02**	99.80±0.10	99.70±0.06	99.80±0.06
OA (%)	89.88± 1.35	90.69±1.02	91.58±1.38	94.02±0.57	94.62±0.83	95.36±1.34	95.81±0.66	96.63±1.24	96.55±0.65	97.21±0.14	97.26±0.55	97.37±0.15	**97.72±0.16**
AA (%)	84.72± 1.49	97.20±0.27	80.56±5.70	89.95±1.20	91.32±1.24	90.70±3.34	93.23±1.23	92.18±2.14	93.20±1.05	93.45±0.31	94.44±0.98	94.53±0.31	**95.27±0.53**
κ*100	88.11± 1.64	94.19±0.52	90.13±1.63	93.00±0.66	93.69±0.98	94.57±1.57	95.66±0.78	96.05±1.46	95.97±0.76	96.74±0.16	96.79±0.65	96.92±0.18	**97.34±0.19**

overall accuracy (OA), average accuracy (AA), and Kappa coefficient (κ).

2) Configuration: We implemented the proposed network on both MindSpore and PyTorch frameworks and conducted experiments on a high-performance computing server equipped with an Intel Xeon Silver 4210R processor, 64-GB RAM, and an NVIDIA Quadro RTX 5000 16-GB GPU to ensure efficient model training and inference computations.

3) Comparative Experiments With Other Excellent Networks: To highlight the superior performance of our proposed

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TABLE XIII

DETAILED CLASSIFICATION ACCURACY OF XUZHOU ON DIFFERENT DL-BASED NETWORKS. THE BEST RESULTS ARE OUTLINED IN BOLD

Class No.	ResNet	ContextualNet	SSRN	SSFTT	LSFAT	MATNet	SANet	SS-MSDCNet	MSFI-CNet	IGroupSS-Mamba	DBMGNet	CSCANet	DSW-LAN
1	98.04±3.35	99.22±0.82	98.94±1.14	97.63±0.74	99.29±0.32	99.10±0.58	**99.69±0.12**	98.25±0.25	98.07±0.39	98.99±0.18	98.29±0.43	98.50±0.46	98.49±0.22
2	99.10±0.58	99.67±0.74	99.82±0.45	98.76±0.90	99.11±1.23	99.83±0.16	**99.99±0.02**	99.46±0.34	99.80±0.11	99.34±0.24	99.40±0.34	99.52±0.13	**99.65±0.18**
3	91.22±8.80	90.73±5.80	92.12±5.63	96.40±2.19	95.49±3.09	95.36±4.75	97.50±2.42	96.83±1.37	97.76±1.60	**99.04±0.31**	98.18±1.69	98.74±0.58	98.67±0.42
4	92.73±5.15	96.82±2.34	96.97±1.85	97.93±0.81	98.05±1.02	96.94±4.00	98.32±0.94	98.01±1.26	98.48±0.48	96.81±1.83	98.65±0.92	98.94±0.41	**99.05±0.46**
5	95.69±1.68	95.45±2.57	96.85±2.10	98.71±0.63	98.04±0.81	97.98±0.81	97.67±0.80	99.49±0.25	**99.58±0.13**	98.94±0.19	99.46± 0.14	99.30±0.27	99.44± 0.08
6	89.73±8.37	90.99±6.55	95.19±4.82	97.74±1.38	97.05±1.48	96.40±1.48	97.30±1.31	**99.84±0.28**	99.15±0.41	99.50±0.27	99.26±0.32	99.09±0.49	98.91±0.44
7	93.87±4.37	91.66±10.63	95.82±2.79	98.30±1.12	97.61±0.89	97.42±2.05	97.11±1.42	98.75±0.26	98.51±0.47	98.75±0.25	98.61±0.38	98.75±0.42	**98.83±0.14**
8	97.26±1.83	98.77±0.90	98.54±1.20	98.20±1.27	98.79±0.73	98.86±0.83	98.92±0.82	99.43±0.48	**99.96±0.05**	99.73±0.22	99.92±0.10	99.93±0.11	99.75±0.30
9	98.25±1.32	97.68±2.71	99.43±0.50	94.24±2.14	98.22±1.01	98.20±0.92	**99.30±0.50**	95.98±0.83	96.12±0.52	96.43±0.50	96.34±1.13	96.18±1.31	96.47±0.19
OA (%)	95.94±1.10	96.48±2.56	97.62±0.92	97.84±0.43	98.45±0.33	98.24±0.67	98.68±0.21	98.57±0.16	98.61±0.21	98.77±0.14	98.70±0.32	98.80±0.13	**98.84±0.06**
AA (%)	95.17± 1.21	95.67±2.46	97.08±1.15	97.55±0.42	97.96±0.53	97.79±0.99	98.42±0.37	98.45±0.23	98.60±0.28	98.61±0.22	98.68±0.37	98.77±0.16	**98.81±0.05**
κ*100	94.84± 1.43	95.57±3.16	96.99±1.18	97.26±0.55	98.03±0.42	97.77±0.85	98.33±0.27	98.19±0.20	98.24±0.27	98.44±0.18	98.35±0.41	98.47±0.16	**98.52±0.08**

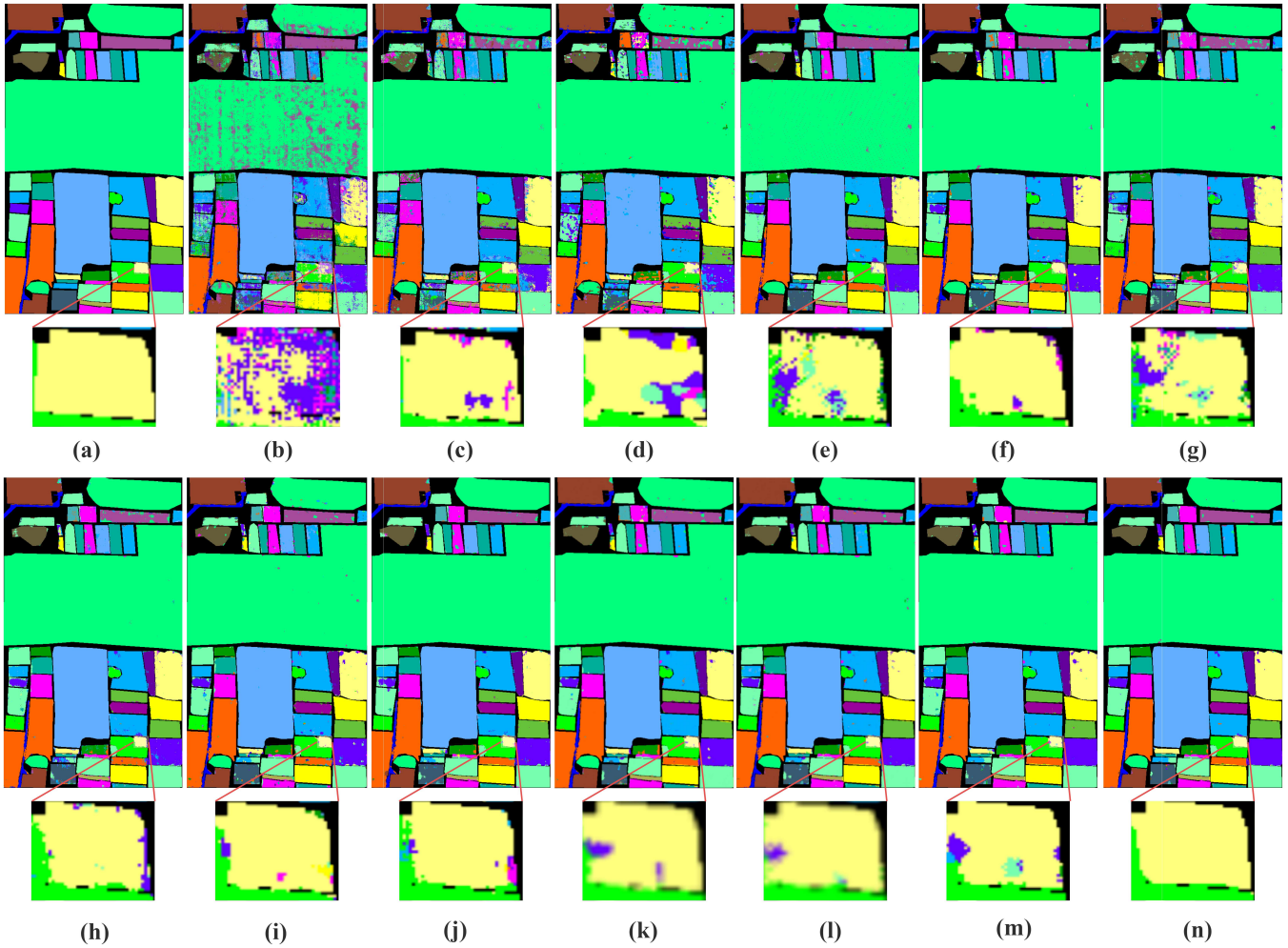


Fig. 8. Visualization of different DL-based networks on WHU-Hi-HongHu. (a) Ground truth. (b) Resnet [51]. (c) ContextualNet [52]. (d) SSRN [53]. (e) SSFTT [54]. (f) LSFAT [55]. (g) MATNet [56]. (h) SANet [57]. (i) SS-MSDCNet [58]. (j) MSFI-CNet [59]. (k) IGroupSS-Mamba [60]. (l) DBMGNet [48]. (m) CSCANet [44]. (n) Proposed DSW-LAN.

network, we selected several state-of-the-art models for comparison, including Resnet [51], ContextualNet [52], SSRN [53], SSFTT [54], LSFAT [55], MATNet [56], SANet [57], SS-MSDCNet [58], MSFI-CNet [59], IGroupSS-Mamba [60], DBMGNet [48], CSCANet [44], and our proposed DSW-LAN.

1) For Resnet [51], ContextualNet [52], SSRN [53], SSFTT [54], LSFAT [55], MATNet [56], SANet [57], SS-MSDCNet [58], MSFI-CNet [59], IGroupSS-Mamba [60], DBMGNet [48], and CSCANet [44], we maintained consistent parameter settings with the original papers.
2) For all models utilizing PCA for dimensionality reduction, we unified the number of reduced spectral channels to 30, ensuring fair evaluation of different models' capabilities in spatial–spectral feature extraction.

4) Experimental Setting:

1) *Using PCA to Reduce Dimension:* Due to the high redundancy of spectral bands in HSIs, we need to per-

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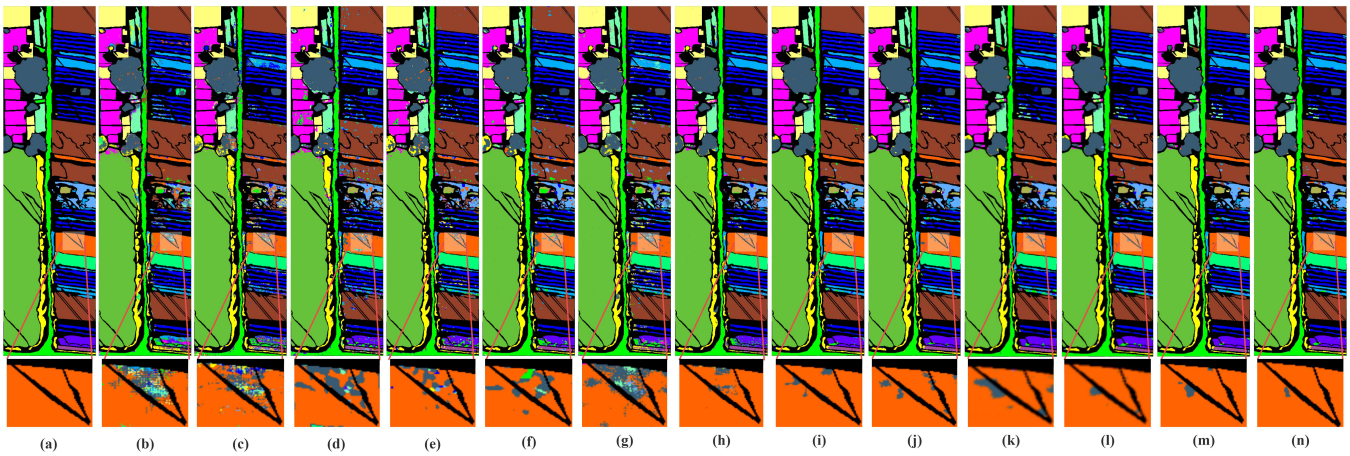


Fig. 9. Visualization of different DL-based networks on WHU-Hi-HanChuan. (a) Ground truth. (b) Resnet [51]. (c) ContextualNet [52]. (d) SSRN [53]. (e) SSFTT [54]. (f) LSFAT [55]. (g) MATNet [56]. (h) SANet [57]. (i) SS-MSDCNet [58]. (j) MSFI-CNet [59]. (k) IGroupSS-Mamba [60]. (l) DBMGNet [48]. (m) CSCANet [44]. (n) Proposed DSW-LAN.

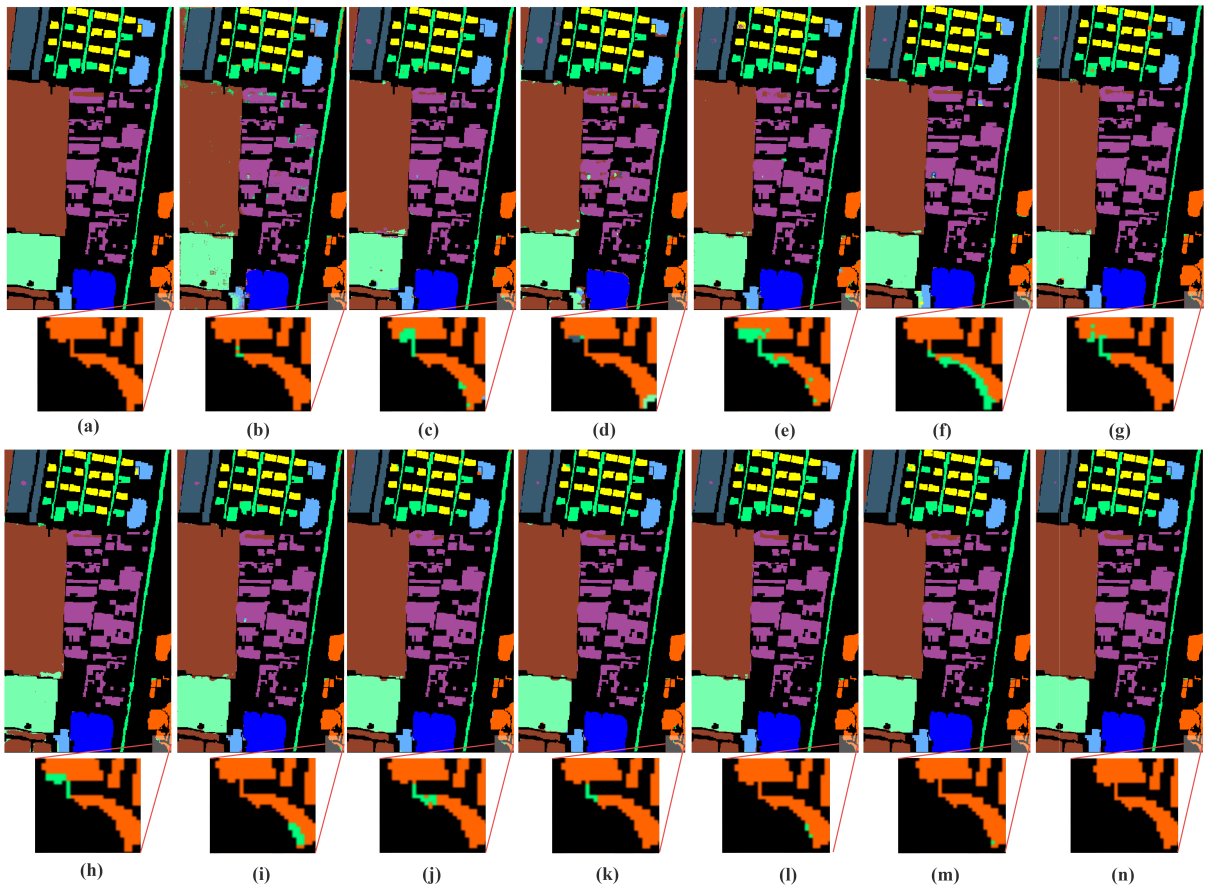


Fig. 10. Visualization of different DL-based networks on XuZhou. (a) Ground truth. (b) Resnet [51]. (c) ContextualNet [52]. (d) SSRN [53]. (e) SSFTT [54]. (f) LSFAT [55]. (g) MATNet [56]. (h) SANet [57]. (i) SS-MSDCNet [58]. (j) MSFI-CNet [59]. (k) IGroupSS-Mamba [60]. (l) DBMGNet [48]. (m) CSCANet [44]. (n) Proposed DSW-LAN.

form dimensionality reduction to reduce computational resource consumption and ensure model operation. We set the number of reduced bands to 30.

2) *Setting the Number of Epochs:* We generated loss curves during training, revealing that the loss stabilizes at a minimum of 0.0001 when the epoch reaches 150 and 250, as shown in Figs. 5 and 6. Furthermore, our observations indicate that if the epoch is set below these values, the loss may stabilize while still being capable of further decline, resulting in slightly lower overall classification accuracy compared to when the loss stabilizes at 0.0001. Consequently, we selected 150 and 250 as the optimal training epochs.

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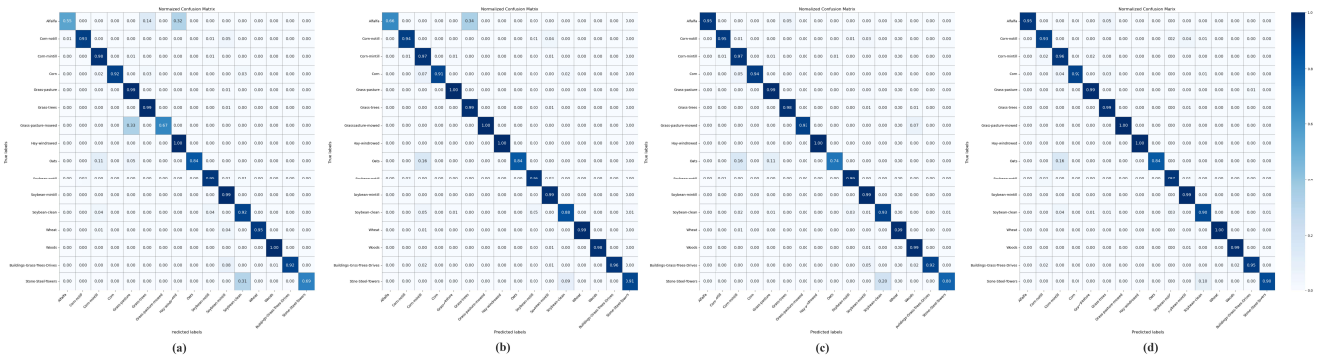


Fig. 11. Confusion matrices visualization of different networks on Indian Pines. (a) SSFTT. (b) MSFI-CNet. (c) CSCANet. (d) Proposed DSW-LAN.

3) *Setting the Size of Factorized Kernel:* When designing the DFDWConv3D block, we adopt $3 \times 3 \times 3$ cubic convolution kernels to maintain consistency with established architectures like MATNet, SANet, SSFTT, and CSCANet. For the directionally decomposed long kernels across spatial dimensions, we conducted comparative experiments on the WHU-Hi-HanChuan dataset. The kernel sizes were configured as [7, 9, 11, 13]. This configuration is based on two considerations. The kernels must be large enough to effectively capture long-range contextual relationships. In addition, constrained by the $13 \times 13 \times 30$ dimensions of the feature cuboid, the maximum kernel size is limited to 13 in spatial domains to maintain valid convolution boundaries. The comparative experimental results are detailed in Table III. We found that when setting the long kernel size to 11, the experimental results achieved the optimal performance. As the kernel size increases to 13, it may cause excessive smoothing of some detailed features while also increasing both the model's training parameters and computational load. Furthermore, the network becomes more prone to overfitting, leading to a significant reduction in classification accuracy.
4) *Setting the Level of Wavelet Transform:* We conduct a detailed study on the optimal value of the level of wavelet transform, and the results are shown in Table IV. The results show that when the level of wavelet transform is set to 3, the classification effect performs optimally. Increasing the level of wavelet transform does not have a significant impact on the experimental results but instead increases the time cost of training the model.
5) *Setting the Number of Spatial and Spectral Feature Enhancement:* Finally, we conducted comparative experiments on the quantity of spatial and spectral feature enhancement modules. The experimental results, as detailed in Table V, demonstrate that setting it to two achieves optimal performance. Single-layer struggles to capture multilevel, progressive spatial and spectral features, resulting in the model being unable to effectively discern subtle distinctions. While two layers prove sufficient to capture essential feature representations, additional layers conversely introduce parameter redundancy, leading to increased optimization difficulties for the model.

5) Comparison Experiment:

1) *Effectiveness Comparison of DFDWConv3D:* To evaluate the effectiveness of our DFDWConv3D, we perform comparative experiments on six other outstanding attention methods, including CBAM [61], SimAM [62], EMA [63], CAA [64], LiteMLA [65], and CSCA [44]. Experimental validation was conducted on the WHU-Hi-HanChuan dataset through systematic module replacement, where our DFDWConv3D was replaced with each comparative attention mechanism while maintaining identical network configurations. As demonstrated in Table VI, the proposed DFDWConv3D achieves superior classification accuracy compared to all alternative attention approaches.
1) *Effectiveness Comparison of SLA:* Considering the structural resemblance between our spectral feature enhancement component and Transformer architectures, we substituted it with advanced Transformer modules exhibiting exceptional performance. We perform comparative experiments on four other outstanding Transformer methods, including morphFormer [33], SPTs [66], DSFormer [67], and MamTrans [68]. Experimental validation was conducted on the WHU-Hi-HanChuan dataset through systematic module replacement, where our DFDWConv3D was replaced with each comparative attention mechanism while maintaining identical network configurations. As demonstrated in Table VII, the proposed SLA achieves superior classification accuracy compared to other Transformer-based approaches.

C. Ablation Study

We conduct comprehensive ablation studies to validate our method and add an efficiency analysis at the end of Table VIII.

1) Effectiveness of DFDWConv3D Block: To evaluate the effectiveness of our designed DFDWConv block, we conducted sufficient experiments comparing results with and without this component. As shown in Tables VIII and IX, the result with DFDWConv block achieves superior classification accuracy with significantly reduced standard deviation,

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demonstrating that our direction-factorized DWConv3D design effectively enhances model robustness.

2) Effectiveness of Wavelet Transform: A systematic evaluation of the wavelet transform component was conducted by comparing model configurations with and without this module. As detailed in Tables VIII and IX, the results demonstrate that the wavelet transform-integrated model achieves the best performance.

3) Effectiveness of SLA: To validate the enhanced discriminative power of SLA in feature learning, we conduct experiments on strategy without using SLA on two datasets. As evidenced in Tables VIII and IX, our proposed SLA module effectively focuses on discriminative features in different spectral bands.

D. Classification Results and Visualization

1) Classification Results: We have presented comprehensive classification results of our method on three datasets, as shown in Tables X–XIII. We can see clearly from the tables that our proposed DSW-LAN achieves state-of-the-art overall accuracy (OA) with minimal variance over ten trials. Moreover, our proposed method achieves optimal accuracy on most LC classes in all three datasets. Overall, these multidimensional evaluations confirm that our proposed method performs particularly well and is highly stable in HSI classification.

Furthermore, to further validate the efficiency of our method, we compared the total parameters, FLOPs, FPS, and memory requirements of different models in Table XI. As FPS directly reflects inference speed under identical training conditions, higher FPS indicates faster model execution. Although our method does not achieve the lowest parameter count, it outperforms most current deep learning-based approaches and attains an optimal tradeoff between classification accuracy and computational efficiency.

2) Visual Results: To provide a clearer demonstration of our method's superior classification performance, we visualized the classification results on three datasets, as shown in Figs. 7–10. The visualizations further indicate that DSW-LAN outperforms other network models, particularly in accurately classifying pixels along the boundaries of different LC classes. In addition, we zoomed in on the areas prone to misclassification in Fig. 9, which provides additional evidence beyond accuracy metrics.

Furthermore, to demonstrate our model's accuracy in classifying similar crop categories, Fig. 11 presents the confusion matrices comparing our model with several other spectral–spatial feature extraction networks. As clearly shown in the figure, our confusion matrix exhibits stronger concentration along the diagonal region with darker color intensity, indicating the superior performance of our model in distinguishing similar crop types.

In the successful cases, the model effectively distinguished texture differences among specific crops in certain areas, benefiting from the critical local patterns and spectral features extracted by our model. For instance, when distinguishing similar crop types such as alfalfa and corn, the DFDWConv3D effectively extracts key spatial and spectral features that reflect surface texture variations and capture spectral responses associated with changes in growth stages. This detailed feature extraction significantly improves the classification accuracy in complex agricultural environments. Furthermore, the wavelet transform Conv enhances this process by separating coarse structural elements from fine details, which is particularly beneficial for differentiating between crops like soybean and oats that exhibit overlapping characteristics. Additionally, the SLA dynamically amplifies significant spectral bands while suppressing irrelevant ones, enabling the model to focus on the most discriminative features for crops such as wheat and grass. This cohesive approach ensures robust classification performance across diverse agricultural scenarios.

However, in some less favorable cases, we found that the model's performance was constrained by insufficient sample sizes. For certain categories like oats, the lack of adequate representative samples hindered the model's ability to effectively learn their features, resulting in decreased classification performance. This finding underscores the importance of ensuring that each category has sufficient samples in agricultural HSI classification tasks to enhance the overall reliability and accuracy of the model.

V. Conclusion

We developed a dual-stage wavelet and linear attention enhancement network for HSI classification, addressing the limitations of conventional methods. Our network uses a DFDWConv3D block for multiscale feature extraction, transforms features into the frequency domain with wavelet-based convolution, and employs a spectral-probe-guided linear attention mechanism to capture global spectral dependencies while reducing computation.

Improved classification accuracy in HSI analysis has significant practical implications for agriculture. Enhanced precision not only optimizes resource allocation—such as water, fertilizers, and pesticides—but also maximizes yield and sustainability. Furthermore, accurate classification supports precision agriculture by facilitating targeted management practices and enhancing decision-making processes.

Despite these advancements, our method has limitations in few-shot learning and struggles with scarce sample categories and anisotropic patterns like urban grids.

Future work will focus on improving few-shot learning through augmented methods, exploring deformable wavelet bases and multidirectional filtering for anisotropic patterns, using data augmentation for synthetic sample generation, and optimizing the model via quantization and pruning to reduce size and computing demands without sacrificing accuracy.

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